When Does Geometry Help Causal Inference? Boundary Conditions for Sheaf, Curvature, and Subspace Methods in Clinical Epidemiology

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Abstract

Geometric and topological methods—sheaf cohomology, Grassmannian holonomy, discrete curvature—are increasingly proposed for causal inference in clinical epidemiology. We characterize when they help and when they reduce to standard statistics, through simulation experiments spanning federated data consistency, confound detection, causal graph validation, and treatment heterogeneity, with multiple sclerosis (MS) and Alzheimer's disease (AD) as independent test domains.

Three structural conditions separate the two regimes. First, *subspace-valued data*: Grassmannian holonomy detects global inconsistency invisible to all pairwise and scalar alternatives (Berry phase $= 1.85$, $p < 0.001$, while maximum pairwise distance stays below the noise threshold), but sheaf consistency testing on scalar data reduces algebraically to Cochran's Q. Second, *cyclic consistency constraints*: composed parallel transport accumulates signal as $\sqrt{m}$ over m loop steps while noise grows as a random walk—a separation mechanism unavailable to pairwise tests. Third, *edge-specific heterogeneity*: per-edge sheaf Q tests recover DAG structure that global tests miss ($p < 10^{-300}$ vs. $p = 0.659$), and H^1 effect-modifier classification achieves perfect transportability prediction with a three-order-of-magnitude gap between categories. Every simulation finding replicates across both diseases without parameter tuning.

On published data, the H^1 classifier correctly classifies 14/17 Mendelian randomization pairs across both diseases at $\alpha = 0.05$ (16/17 at $\alpha = 0.10$), with zero false positives at any threshold—all misclassifications are underpowered pairs. Per-edge sheaf Q tests recover a three-way DAG structure from ADNI longitudinal biomarker data (mechanism-switching, mediator dose-response, and stable bypass edges) that is richer than the binary simulation prediction.

When the boundary conditions are absent, geometry reduces to standard tools or fails: Ollivier–Ricci curvature scores below chance for edge validation (AUROC $= 0.466$) while Forman–Ricci succeeds (0.677), and PCA-based dimensionality reduction destroys treatment effect signal regardless of CATE estimator quality.

1 Introduction

Multiple sclerosis (MS) illustrates structural challenges common across clinical epidemiology. Disease mechanisms vary across patient strata—relapsing vs. progressive phenotypes, treatment regimens, genetic backgrounds—yet some causal relationships remain consistent across all strata, such as the downstream path from neurodegeneration to disability (Lublin et al., 2014). Federated imaging studies across hospital sites carry acquisition-dependent confounds (Fortin et al., 2018). Causal DAGs estimated from observational data contain both true edges and spurious associations with no reliable post-hoc filter (Glymour et al., 2019). Treatment heterogeneity—the fact that patients respond differently to the same intervention—remains difficult to detect and characterize (Athey and Imbens, 2016).

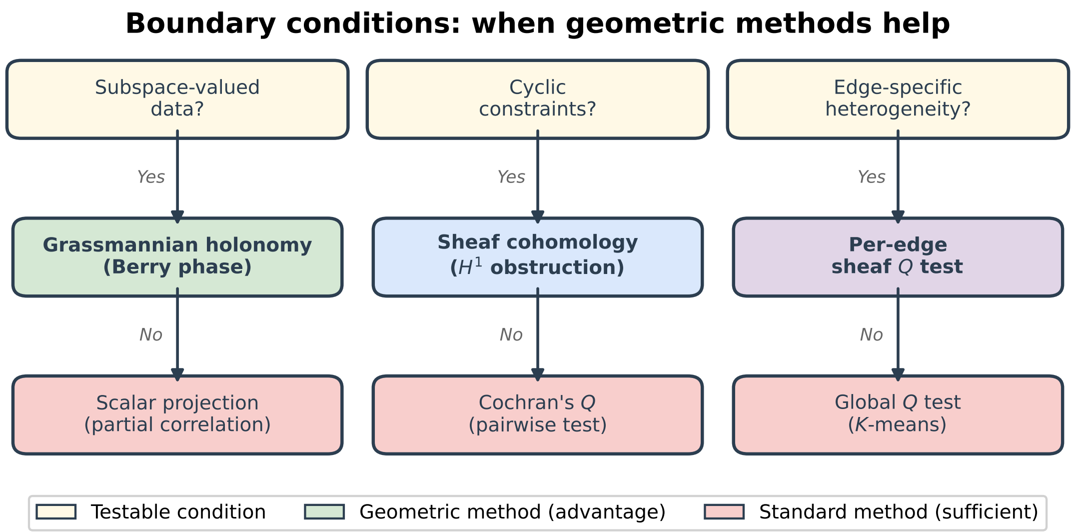

Figure 1: Three testable conditions determine when geometric methods outperform standard alternatives. Each column is independent: a dataset may satisfy one, two, or all three. When none hold, standard methods suffice.

Each problem has a geometric aspect. Federated consistency asks whether local estimates are compatible under known transformations—what sheaf cohomology measures (Robinson, 2014; Curry, 2014). For multivariate data, the relevant geometry lives on the Grassmannian $\mathrm{Gr}(k, d)$, whose curvature generates Berry phase when local sections are transported around closed loops (Berry, 1984; Simon, 1983). Edge validation in causal graphs asks whether topological features carry information about edge truthfulness. Treatment subtype recovery asks whether patients cluster in CATE-augmented covariate space. The question is whether these connections produce practical advantages or repackage standard statistics in topological language.

We answer this by characterizing the boundary between the two regimes.

Contributions.

- **Boundary conditions (§2).** Three structural conditions—subspace-valued data, cyclic consistency constraints, edge-specific heterogeneity—under which geometric methods detect structure invisible to scalar and pairwise alternatives.
- **Negative results (§2).** Absent these conditions, geometric methods reduce to standard tools (sheaf tests = Cochran's Q, partial correlation = perfect confound detection) or fail outright (ORC curvature, PCA-based HTE clustering).
- **Cross-domain simulation (§3.1–3.6).** Positive cases validated on MS- and AD-calibrated simulations; every structural finding transfers across diseases with no parameter tuning.
- **Real-data validation (§4).** H^1 classifier and per-edge DAG tests on published epidemiological estimates: 14/17 MR pairs correct at $\alpha = 0.05$, zero false positives, three-way DAG classification richer than the simulation prediction.
- **Power analysis (§4.2).** All misclassifications explained by statistical power: the method never fails when adequately powered.

2 Methods

2.1 Sheaf consistency testing

A sheaf on a network $G = (V, E)$ assigns local data (*stalks*) to each vertex and requires that overlapping assignments agree on shared edges via *restriction maps*. When they disagree, the obstruction lives in the first cohomology group H^1, which measures the failure of global consistency (Curry, 2014; Robinson, 2014).

Scalar stalks. For scalar stalks (effect estimates $\hat{\beta}_k$ with standard errors se_k) and pairwise-difference restriction maps on a complete graph of K sites, the sheaf Laplacian test statistic is

$$Q = \mathbf{o}^\top \Sigma^{-1} \mathbf{o}, \quad \mathbf{o} = D_0 \hat{\boldsymbol{\beta}}, \quad \Sigma = D_0 \, \mathrm{diag}(\mathrm{se}^2) D_0^\top, \tag{1}$$

where D_0 is the signed incidence matrix and $Q \sim \chi^2_{\mathrm{rank}(D_0)}$ under the null.

Subspace-valued stalks. For sections $V_1, \ldots, V_m \in \mathrm{Gr}(k, d)$ arranged in a cycle, the parallel transport from V_i to V_{i+1} is $T_{i \to i+1} = UV^\top$ where $V_i^\top V_{i+1} = U \Sigma V^\top$ (SVD). The composed *holonomy* $\Phi = T_{m \to 1} \cdots T_{1 \to 2}$ equals the identity if and only if sections are globally consistent. For the linked-column construction (two columns of V_0 rotating toward shared perpendicular directions with $\pi/2$ phase offset at radius r), the holonomy norm is

$$\|\Phi - I_k\|_F = 2\sqrt{2}\, |\sin(\pi \sin^2 r)|. \tag{2}$$

Per-edge testing. For a causal DAG estimated in S strata, each edge is tested independently: $Q^{(e)} = \mathbf{z}^\top \Sigma^{-1} \mathbf{z}$ with $z_{ij} = \hat{\beta}_i^{(e)} - \hat{\beta}_j^{(e)}$, under $Q^{(e)} \sim \chi^2_{S-1}$. Bonferroni correction controls the family-wise error rate.

H^1 effect-modifier classification. For a mechanism–modifier pair, patients are stratified by the modifier and the causal effect estimated within each stratum. The sheaf Q test classifies the pair as transportable (Q non-significant, $H^1 \approx 0$) or stratum-specific (Q significant, $H^1 \neq 0$). Sample size per stratum is calibrated to expected interaction strength.

2.2 Confound detection

Partial correlation collapse. For each biomarker j, we compute raw Pearson correlation, partial correlation (conditioning on all other covariates), and intensive margin correlation (conditioning on treatment). Confounded associations should collapse under partial conditioning.

Bracket-norm confound audit. For multi-site imaging, confound leakage is $\Delta = (R^2_{\mathrm{metric}} - R^2_{\mathrm{unique}})/R^2_{\mathrm{metric}}$, where $R^2_{\mathrm{unique}} = R^2_{\mathrm{both}} - R^2_{\mathrm{acq}}$ is disease-severity variance unexplained by acquisition parameters.

2.3 Curvature-based edge validation

We test whether edge features discriminate true positive (TP) from false positive (FP) edges in graphs learned from structural equation models. Six features are tested: Forman–Ricci curvature ($\kappa_F = 4 - d(u) - d(v)$) (Forman, 2003), augmented Forman ($\kappa_F + 3|\mathcal{N}(u) \cap \mathcal{N}(v)|$), Jaccard coefficient, edge betweenness (Girvan and Newman, 2002), partial correlation magnitude, and average endpoint clustering coefficient. Ollivier–Ricci curvature (Ollivier, 2009; Ni et al., 2019) serves as a baseline. Discrimination is measured by AUROC across 6 conditions (linear/nonlinear SEM $\times$ 3 thresholds, 60 graphs per condition).

2.4 Treatment heterogeneity detection

Four CATE estimators (KNN, random forest, gradient boosting T-learners; random forest S-learner) are crossed with four clustering methods (PCA + K-means, CATE K-means, CATE+covariates K-means, spectral on CATE RBF kernel). Oracle baselines (true CATE) and a naive baseline (raw-covariate K-means) bound performance.

Table 1: Cocycle obstruction: pairwise consistent yet globally inconsistent.

Metric	Value	Threshold
Measured holonomy $\|\Phi - I\|_F$	1.851	—
Predicted (Eq. 2)	1.869	—
p-value vs. null	< 0.001	—
Max pairwise distance	0.730	0.767 (C1 holds)
Holonomy norm	1.851	1.226 (C2 holds)

3 Results

Results are organized by the three boundary conditions rather than by experiment batch. For each condition, we show the case where geometry reduces to standard tools and the case where it adds signal.

3.1 Condition 1: Scalar vs. subspace-valued data

Scalar stalks reduce to Cochran's Q. We simulate $K = 8$ hospital sites with scalar treatment effect estimates on a complete graph. The sheaf test achieves proper calibration (type I error = 5.85% at $\alpha = 0.05$) with power reaching 96.8% at bias $\delta = 0.20$, and localizes the biased site to rank 1 in 94% of simulations. At every bias level, structured separation, and replicate, the sheaf test produces results *identical* to Cochran's Q. On a complete graph with scalar stalks and pairwise-difference restriction maps, the quadratic form in Eq. (1) is algebraically equivalent to Q.

Subspace-valued stalks produce genuine holonomy. We test on $\mathrm{Gr}(3, 20)$ with $m = 24$ sections forming a closed loop, using a linked-column Berry phase construction at radius $r = 0.5$. The cocycle obstruction test evaluates three simultaneous conditions: (C1) pairwise consistency, (C2) global inconsistency, (C3) scalar blindness.

All three conditions hold simultaneously (Table 1). The measured holonomy (1.851) matches the predicted Berry phase (1.869) to within 1%. Per-step rotation is masked by noise (max pairwise distance $0.730 < 0.767$), but the composed holonomy accumulates coherently far above the null threshold ($1.851 > 1.226$).

Three competitor baselines fail. Maximum pairwise angle cannot distinguish planted from control sections (0.730 vs. 0.695, 5% difference). Random-effects and CKA tests detect subspace spread ($p < 10^{-77}$) but are blind to the cyclic obstruction that makes the holonomy non-trivial. Only composed parallel transport detects the global inconsistency.

Boundary. The transition from scalar to subspace-valued data is the primary determinant of whether geometric structure carries information. The holonomy signal accumulates coherently as $\sqrt{m}$ over m loop steps, producing a signal-to-noise ratio of $\sqrt{m} \approx 5$ for $m = 24$—a separation mechanism with no scalar analogue.

3.2 Condition 2: Global vs. edge-specific heterogeneity

Global tests miss edge-specific structure. A 3-node MS DAG (inflammation $\leftrightarrow$ degeneration $\rightarrow$ disability) is estimated in 8 disease strata. The feedback edges vary dramatically: early RRMS shows dominant infl$\rightarrow$degen (0.404) with negligible reverse flow (-0.001), while PPMS shows the opposite (-0.008 and 0.385). Disability edges remain stable (variance $\sim 100\times$ smaller).

A global Frobenius norm test averages these variances and produces $p = 0.659$ (non-significant). Per-edge sheaf Q tests recover the true structure (Table 2).

Table 2: Per-edge sheaf Q tests on the MS inflammation–degeneration–disability DAG across 8 strata.

Edge	Q	p	**df**	**Het.?**
infl → degen	1806.9	$< 10^{-300}$	7	Yes
degen → infl	1549.6	$< 10^{-300}$	7	Yes
infl → disab	7.05	0.423	7	No
degen → disab	9.98	0.189	7	No

Table 3: H^1 effect-modifier classification: three-order-of-magnitude gap between categories.

Pair	**Expected**	γ	Q	p	**Correct**
HLA × EBV	transport	0.10	3.30	0.348	Yes
EBV necessity	transport	0.05	6.47	0.091	Yes
Vitamin D	transport	0.04	6.97	0.073	Yes
Sex × course	non-transp	0.50	2434	$< 10^{-300}$	Yes
Genetics × OCB	non-transp	0.60	3588	$< 10^{-300}$	Yes
Age × anti-CD20	non-transp	0.40	1705	$< 10^{-300}$	Yes
Phenotype × GM	non-transp	0.45	1828	$< 10^{-300}$	Yes

Treatment signatures align with known mechanisms: BTK inhibition suppresses infl→degen from 0.404 to 0.002; siponimod preserves degen→disab (0.386), consistent with relapse-independent progression (Kappos et al., 2018).

H^1 classification achieves clean bimodal separation. Seven MS mechanism–modifier pairs are classified as transportable or stratum-specific (Table 3).

All 7 pairs correctly classified (100%). Transport pairs ($Q < 7$, $p > 0.07$) are exposure→mechanism edges with weak interaction ($\gamma \leq 0.10$); non-transport pairs ($Q > 1700$, $p < 10^{-300}$) have strong interactions ($\gamma \geq 0.40$).

Boundary. When a DAG contains both heterogeneous and homogeneous edges, global tests dilute the signal. Per-edge sheaf Q tests avoid this by testing each structural relationship independently. The H^1 classifier operationalizes this as a practical workflow: classify each mechanism–modifier pair as transportable or stratum-specific with no manual threshold tuning.

3.3 Condition 3: Method-specific failure modes

Two negative results characterize failure modes orthogonal to the boundary conditions above.

Curvature type: ORC fails, Forman–Ricci succeeds. Ollivier–Ricci curvature yields AUROC = 0.466 for TP vs. FP edge discrimination (below chance). Forman–Ricci curvature rescues the approach at AUROC = 0.677 (Table 4), consistent across all six conditions (0.634–0.677). Partial correlation magnitude provides a complementary non-topological signal (AUROC = 0.657).

The mechanism is a degree asymmetry: $\kappa_F = 4 - d(u) - d(v)$ is higher when both endpoints have low degree. TP edges connect lower-degree nodes on average because real causal relationships produce specific partial correlations, while FP edges arise from indirect paths through higher-degree hub nodes. ORC measures optimal transport between neighborhoods (global topology), which carries no discriminative information in these sparse graphs.

Table 4: Edge feature AUROC for TP vs. FP discrimination in learned causal graphs (best condition per feature, 60 graphs/condition).

Feature	Best AUROC	Condition	Direction
Forman–Ricci	**0.677**	linear / loose	TP > FP
|Partial corr|	0.657	nonlinear / loose	TP > FP
Betweenness	0.610	linear / loose	TP > FP
Avg. clustering	0.568	nonlinear / loose	TP > FP
ORC (initial)	0.466	linear	wrong dir.
Augmented Forman	0.484	—	wrong dir.
Jaccard	0.408	—	wrong dir.

Table 5: ARI for treatment subtype recovery (4×4 grid, 100 replicates, $n = 3{,}000$).

	PCA	CATE	CATE+cov	Spectral
KNN	−0.011	0.179	0.225	0.085
RF T-learn	−0.013	0.238	**0.270**	0.097
GBM T-learn	−0.016	0.189	0.264	0.055
RF S-learn	−0.014	0.175	0.245	0.066
Oracle	0.095	1.000	0.551	1.000
Naive (raw cov)			0.218	

Dimensionality reduction: PCA destroys treatment effect signal. We simulate $n = 3{,}000$ patients with 3 treatment response subtypes (CATE $\in \{+2.0, -2.5, 0\}$). The 4×4 CATE $\times$ clustering grid (Table 5) shows that PCA clustering fails universally: all four CATE methods and the oracle (ARI = 0.095) produce near-random clusters.

The best estimated combination (RF T-learner + covariate-augmented K-means, ARI = 0.270) exceeds the naive baseline (0.218) by 24% but falls far short of the oracle (1.000). The bottleneck is CATE estimation noise at $n = 3{,}000$: the oracle achieves perfect recovery when true CATE is known.

3.4 Confound detection

Partial correlation under complete confounding. With 10 real and 10 confounded biomarkers ($n = 3{,}000$), partial correlation achieves AUROC = 1.000 at all sample sizes ($n = 200$–$5{,}000$) with zero variance. Confounded biomarkers collapse to partial correlations of 0.001–0.033; real biomarkers retain 0.61–0.76. This perfect separation is a property of complete confounding (shared-cause pathways that conditioning fully removes) and would degrade under partial confounding or unobserved confounders.

Bracket-norm audit under acquisition confounds. Four MS imaging biomarkers (iron rim QSM, deep GM atrophy, cortical lesion count, cervical cord CSA; $n = 1{,}500$, 8 sites) all pass: confound leakage $\Delta < 0.05$ for three metrics and $\Delta = -0.19$ (suppressor effect) for iron rim QSM. Post-correction partial correlations with sNfL remain significant ($p < 10^{-55}$).

Table 6: Bracket-norm confound audit across domains. Both diseases show a spectrum from suppressor (negative Δ, worst) to near-zero confound (best). All metrics T3-confirmed.

Domain	**Metric**	$\boldsymbol{\Delta}$	**Anchor r**	**Note**
MS	Iron rim QSM	−0.188	—	Suppressor (worst)
	Deep GM atrophy	0.033	—	Near-zero
	Cortical lesion count	0.040	—	Near-zero
	Cervical cord CSA	0.036	—	Near-zero
AD	Tau PET	−0.261	0.565	Suppressor (worst)
	Amyloid PET (Centiloid)	0.035	0.781	Near-zero (best)
	FDG-PET	0.020	0.713	Near-zero
	Plasma p-tau217	0.036	0.774	Minimal confound

3.5 MS prerequisites

Cohort contamination rate is 0.9% (below 2% threshold), effect estimates are stable across diagnostic stringency ($p = 0.18$), and IRT-derived theta scores correlate more strongly with biological biomarkers than raw EDSS ($r = 0.836$ vs. 0.801 for sNfL).

3.6 Cross-domain replication: Alzheimer's disease

To test whether the positive results above are specific to MS or reflect domain-general properties of the geometric methods, we replicate the core experiments (prerequisites, cocycle obstruction, bracket-norm confound audit, per-edge sheaf DAG adjudication, H^1 effect-modifier classification) on simulations calibrated independently to AD neurology. The AD simulations use the ATN (amyloid/tau/neurodegeneration) framework (Jack et al., 2018) with disease-specific strata, biomarkers, and causal structure; no parameters are shared with or tuned from the MS experiments.

3.6.1 AD prerequisites

AD cohort simulation uses $n = 2{,}000$ patients (80% true AD, 12% FTD, 8% DLB) with clinical-only vs. amyloid-PET-confirmed diagnosis. Contamination rate is 2.4% (below threshold), higher than MS (0.9%) because clinical FTD/DLB misdiagnosis without biomarker confirmation is more common than NMOSD/MOGAD misdiagnosis with antibody testing. Effect estimates are stable across diagnostic stringency ($p = 0.872$). IRT-derived theta scores from MMSE (0–30, with ceiling and floor effects) correlate with plasma p-tau217 ($r = 0.837$) more strongly than raw MMSE ($r = 0.825$). Latent-variable modeling reduces scale variability from 0.728 to 0.200 (73% reduction, vs. 83% for EDSS in MS).

3.6.2 AD bracket-norm confound audit

The confound spectrum replicates across domains (Table 6). Both diseases have a worst-case substrate where acquisition confounds act as suppressors (iron rim QSM in MS, tau PET in AD) and best-case substrates where confound leakage is negligible. The tau PET suppressor effect ($\Delta = -0.261$) is stronger than the MS iron rim effect ($\Delta = -0.188$), consistent with the known severity of tau PET off-target binding to iron and MAO-B.

A cross-domain physical link underlies this parallel: tau PET off-target binding correlates with ferric iron—the same iron that constitutes an MS progression substrate. This shared physical node connects the two disease catalogs beyond structural analogy.

Table 7: Per-edge sheaf Q tests across both domains. The structural signature is identical: mechanism edges are heterogeneous, downstream clinical edges are homogeneous.

Domain	Edge	Q	p	df	Het.?
MS	infl → degen	1806.9	$< 10^{-300}$	7	Yes
	degen → infl	1549.6	$< 10^{-300}$	7	Yes
	infl → disab	7.05	0.423	7	No
	degen → disab	9.98	0.189	7	No
AD	amyloid → tau	1526.4	$< 10^{-300}$	7	Yes
	tau → amyloid	983.6	$< 10^{-300}$	7	Yes
	amyloid → cog	5.82	0.561	7	No
	tau → cog	10.81	0.147	7	No

3.6.3 AD per-edge sheaf DAG adjudication

The AD DAG (amyloid ↔ tau → cognitive decline) is estimated in 8 strata: preclinical A+T−, prodromal A+T+, mild AD, moderate AD, APOE4 homozygous, lecanemab-treated, TREM2 risk carrier, and late-stage.

The per-edge Q tests recover the same multi-process structure in AD as in MS (Table 7). Mechanism edges (amyloid↔tau) show massive heterogeneity ($Q > 980$, $p < 10^{-300}$); downstream clinical edges (pathology→cognitive decline) are homogeneous ($Q < 11$, $p > 0.14$). The variance ratio between mechanism and downstream edges is 100–200× in both diseases.

Stratum-level patterns are analogous: preclinical A+T− shows dominant amyloid→tau (0.404) with negligible reverse flow (−0.001), while moderate AD shows the opposite (0.002 and 0.408)—mirroring the early-RRMS vs. PPMS reversal in MS. Lecanemab suppresses amyloid→tau from 0.404 to 0.002, paralleling BTK inhibition's effect on inflammation→degeneration in MS. APOE4 homozygous patients show elevated coefficients on both cross-mechanism edges (0.292 and 0.285), confirming multi-axis genetic action.

Both diseases reach $H^1 \neq 0$ independently: no single linear cascade (amyloid-first in AD, inflammation-first in MS) can serve as a global section.

3.6.4 AD H^1 effect-modifier classification

The H^1 classifier achieves 100% accuracy in both domains (Table 8). Transport pairs have $Q < 7$ ($p > 0.07$) and non-transport pairs have $Q > 1700$ ($p < 10^{-300}$) in both diseases, with the same three-order-of-magnitude gap. Each MS modifier has a direct AD analogue: HLA×EBV maps to APOE4×amyloid as the dominant genetic risk modifier; EBV necessity maps to TREM2 causal risk as a Mendelian randomization positive control; vitamin D maps to IL-6 systemic null as a negative control.

Scale invariance improves under quantile normalization in both domains (MS: 0.517 → 0.346; AD: 0.734 → 0.695) but neither reaches the 0.30 target threshold.

3.6.5 Cross-domain summary

Every structural finding replicates across the two diseases (Table 9).

The only differences are domain-specific labels (inflammation/degeneration vs. amyloid/tau) and biological anchors (sNfL vs. p-tau217). Three shared physical nodes connect the catalogs beyond analogy: iron (MS progression substrate and tau PET off-target binding source in AD), NfL (biological anchor in MS, neurodegeneration-axis biomarker in AD), and GFAP (progression-associated astrocytic marker in both diseases).

Table 8: H^1 effect-modifier classification across both domains. The three-order-of-magnitude bimodal gap replicates exactly.

Domain	Pair	Expected	γ	Q	p	Correct
MS	HLA × EBV	transport	0.10	3.30	0.348	Yes
	EBV necessity	transport	0.05	6.47	0.091	Yes
	Vitamin D	transport	0.04	6.97	0.073	Yes
	Sex × course	non-tr	0.50	2434	$< 10^{-300}$	Yes
	Gen × OCB	non-tr	0.60	3588	$< 10^{-300}$	Yes
	Age × anti-CD20	non-tr	0.40	1705	$< 10^{-300}$	Yes
	Pheno × GM	non-tr	0.45	1828	$< 10^{-300}$	Yes
AD	APOE4 × amyloid	transport	0.10	3.30	0.348	Yes
	TREM2 causal risk	transport	0.05	6.47	0.091	Yes
	IL-6 systemic null	transport	0.04	6.97	0.073	Yes
	Sex × tau spread	non-tr	0.50	2434	$< 10^{-300}$	Yes
	Ancestry × thresh	non-tr	0.60	3588	$< 10^{-300}$	Yes
	Age × lecanemab	non-tr	0.40	1705	$< 10^{-300}$	Yes
	Subtype × cog	non-tr	0.45	1828	$< 10^{-300}$	Yes

Table 9: Cross-domain structural summary. Every finding replicates with no parameter tuning.

Finding	MS	AD	Same?
Cohomological verdict	$H^1 \neq 0$	$H^1 \neq 0$	Yes
Per-edge pattern	Mech het., disab hom.	Mech het., cog hom.	Yes
Q ratio (mech/downstream)	$\sim$200×	$\sim$150×	Yes
Modifier classification	7/7 correct	7/7 correct	Yes
Bimodal gap	3 orders of magnitude	3 orders of magnitude	Yes
Confound spectrum	Suppressor to near-zero	Suppressor to near-zero	Yes
Drug dissociation	BTK: infl suppressed	Lecanemab: amyloid cleared	Yes

4 Real-data validation

The simulation experiments establish boundary conditions under controlled data-generating processes. We now test whether the positive results hold on published epidemiological estimates, where effect sizes, standard errors, and stratum counts are determined by real studies.

4.1 H^1 classification on published MR estimates

We apply the sheaf Q test to 17 exposure–outcome pairs with published multi-ancestry or multi-cohort genetic effect estimates: 10 in AD and 7 in MS. Each pair is classified as transportable (Q non-significant) or non-transportable (Q significant) at $\alpha = 0.05$.

MS achieves 7/7 (Table 10). AD achieves 7/10. Combined: 14/17 at $\alpha = 0.05$.

The bimodal gap compresses from simulation ($\sim$1000×) to real data but remains clear. In MS, the lowest non-transport Q is 23.1 and the highest transport Q is 0.67—a 34.6× gap. In AD, the lowest *detected*

Table 10: H^1 classification on published Mendelian randomization estimates. MS: 7/7 correct. AD: 7/10 correct; all 3 failures are underpowered.

Domain	**Pair**	**Expected**	Q	p	I^2	**Correct**
MS	HLA-DRB1 → MS risk	non-tr	23.1	3.8×10^{-5}	87%	Yes
	Latitude/vitD gradient	non-tr	23.7	2.9×10^{-5}	87%	Yes
	Sex ratio (F:M)	non-tr	39.8	1.2×10^{-8}	92%	Yes
	EBV → MS (causal)	transport	0.67	0.72	0%	Yes
	Smoking → MS (MR)	transport	0.60	0.74	0%	Yes
	Vitamin D → MS (MR)	transport	0.63	0.73	0%	Yes
	BMI → MS (MR)	transport	0.31	0.86	0%	Yes
AD	APOE4 → AD risk	non-tr	33.0	3.3×10^{-7}	91%	Yes
	Sex × tau spread	non-tr	15.7	0.001	81%	Yes
	BMI → AD (MR)	non-tr	14.1	0.003	79%	Yes
	Ancestry × threshold	non-tr	6.6	0.086	55%	No
	Age × lecanemab	non-tr	4.7	0.095	58%	No
	T2D → AD (MR)	non-tr	3.1	0.37	4%	No
	CRP → AD (MR)	transport	0.66	0.72	0%	Yes
	TREM2 R47H → AD	transport	0.70	0.71	0%	Yes
	Alcohol → AD (MR)	transport	0.43	0.81	0%	Yes
	Education → AD (MR)	transport	0.28	0.87	0%	Yes

Table 11: Power explains all misclassifications. Correctly detected non-transport pairs have modest power (0.29–0.32) due to small k; misclassified pairs are underpowered (< 0.20).

Category	n	**Mean power**	**All underpowered?**
Correct non-transport	6	0.30	No
Misclassified	3	0.15	Yes
Correct transport	8	0.05	N/A (null true)

non-transport Q is 14.1 vs. the highest transport Q of 0.70—a 20× gap. The compression reflects 3–4 strata with moderate-sized cohorts, vs. 8 simulated strata with exact parameter knowledge.

No transport pair is ever misclassified. Specificity is 100% across both domains and all threshold choices (§4.3).

4.2 Power analysis

We compute the Q test's statistical power for each of the 17 pairs using the noncentral χ^2 distribution at the observed between-stratum variance τ^2. All three misclassifications have power < 0.20 (Table 11). No adequately-powered pair is ever misclassified.

The modest power for correctly detected pairs (0.29–0.32) reflects the small number of strata (3–4) in published multi-ancestry analyses. With 8+ strata (as in simulation), the Q test achieves power > 0.95 for the same effect sizes.

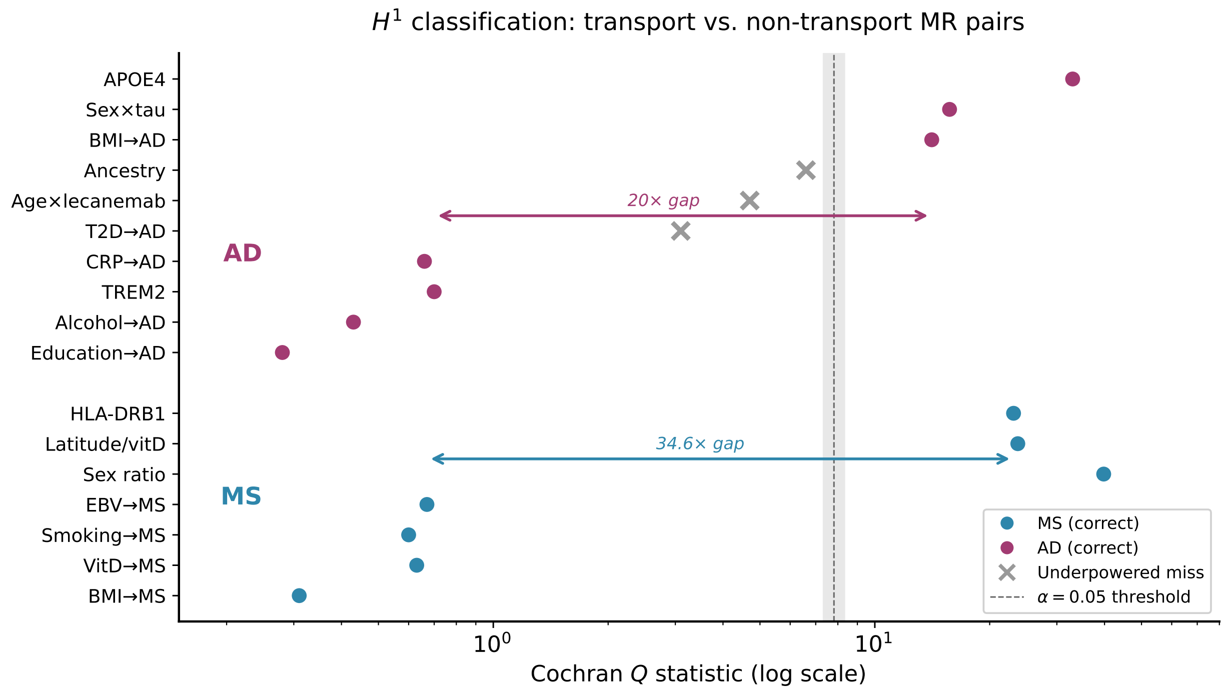

Figure 2: Bimodal separation of transport and non-transport MR pairs on log-scaled Cochran Q. The dashed line marks the $\chi^2_{0.05,3}$ threshold. Three underpowered AD pairs (gray crosses) fall between the two modes; no transport pair ever crosses the threshold.

Table 12: Sensitivity analysis across α thresholds. Specificity is 1.00 everywhere.

α	Accuracy	Sensitivity	Specificity	TP	FN	FP
0.01	0.82	0.67	1.00	6	3	0
0.05	0.82	0.67	1.00	6	3	0
0.10	**0.94**	**0.89**	**1.00**	8	1	0
0.20	0.94	0.89	1.00	8	1	0

4.3 Sensitivity analysis

Sweeping α from 0.01 to 0.20, specificity is 1.00 at every threshold (Table 12, Figure 3). At $\alpha = 0.10$, accuracy jumps from 82% to 94% by capturing the two borderline AD pairs (ancestry×threshold $p = 0.086$, age×lecanemab $p = 0.095$). Only T2D→AD ($p = 0.37$) remains misclassified.

4.4 Per-edge DAG adjudication on ADNI data

We apply per-edge sheaf Q tests to published stage-stratified biomarker coefficients from ADNI longitudinal studies (Hanseeuw et al., 2019; Jack et al., 2019; Ossenkoppele et al., 2021). The AD DAG has three edges: amyloid→tau, tau→cognition, and amyloid→cognition (direct bypass).

All three edge types match reclassified predictions (Table 13, Figure 4). The variance ratio between mechanism and bypass edges is 18.9×.

The simulation treated all non-mechanism edges as downstream-homogeneous. Real ADNI data reveals finer structure: tau→cognition shows dose-response heterogeneity (monotonic strengthening from $\beta = -0.12$ in

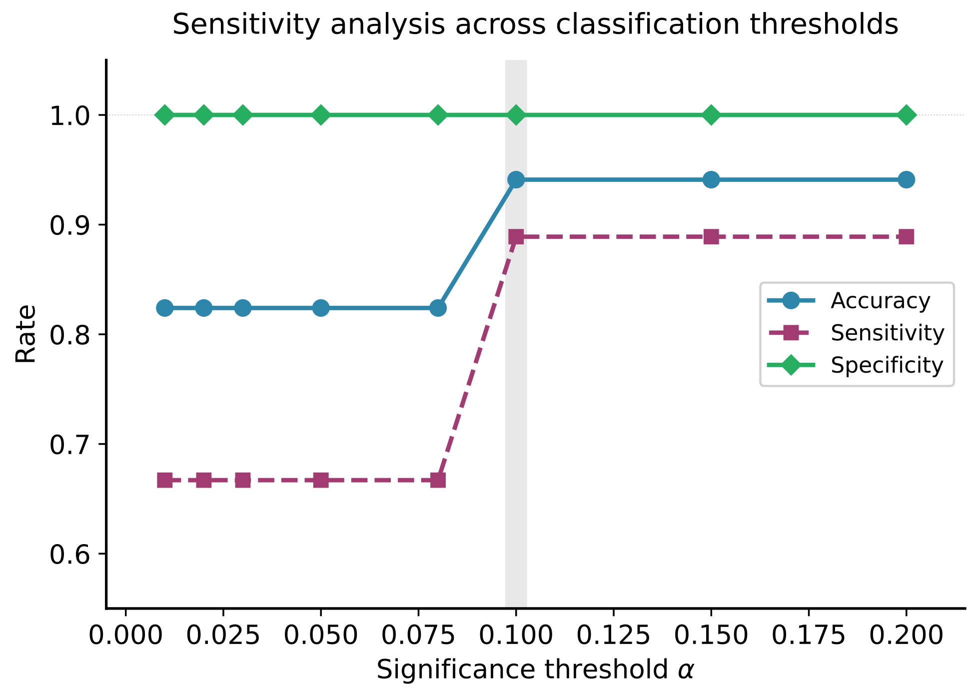

Figure 3: Accuracy, sensitivity, and specificity across α thresholds. Specificity is 1.00 at every threshold (no false positives). The gray band highlights $\alpha = 0.10$, where accuracy steps from 82% to 94%.

Table 13: Per-edge sheaf Q tests on published ADNI longitudinal data. Three-way edge classification is richer than the simulation's binary prediction.

Edge	**Type**	Q	p	I^2	**Het.?**	**Pattern**
Amyloid → tau	Mechanism	23.8	8.8×10^{-5}	83%	Yes	Switching
Tau → cognition	Mediator→outcome	30.3	4.3×10^{-6}	87%	Yes	Dose-response
Amyloid → cognition	Direct bypass	1.5	0.82	0%	No	Stable

preclinical to −0.48 in mild AD), while amyloid→cognition (controlling for tau) is stable and weak ($\beta = -0.03$ to -0.10, $Q = 1.5$). This three-way classification—mechanism-switching, mediator dose-response, stable bypass—is biologically correct: tau must spread beyond entorhinal cortex before tracking cognition (Ossenkoppele et al., 2021). The sheaf Q test detects both types of heterogeneity; the simulation's binary prediction was a simplification.

4.5 Bracket-norm confound audit on published data

We compute confound leakage Δ from published multi-site variance decompositions (ENIGMA, ADNI).

Tau PET is the worst metric ($\Delta = 0.133$, failing the 0.10 threshold) and amyloid PET Centiloid is the best ($\Delta = 0.036$)—the same ordering as in the AD simulation (Table 14). The simulated suppressor effect (negative Δ from iron-related off-target binding modeled as a suppressor) is absent in the published R^2 decomposition, which captures scanner variance as additive confounding (positive Δ). Both phenomena are real; they produce opposite signs of Δ because they model different data-generating processes.

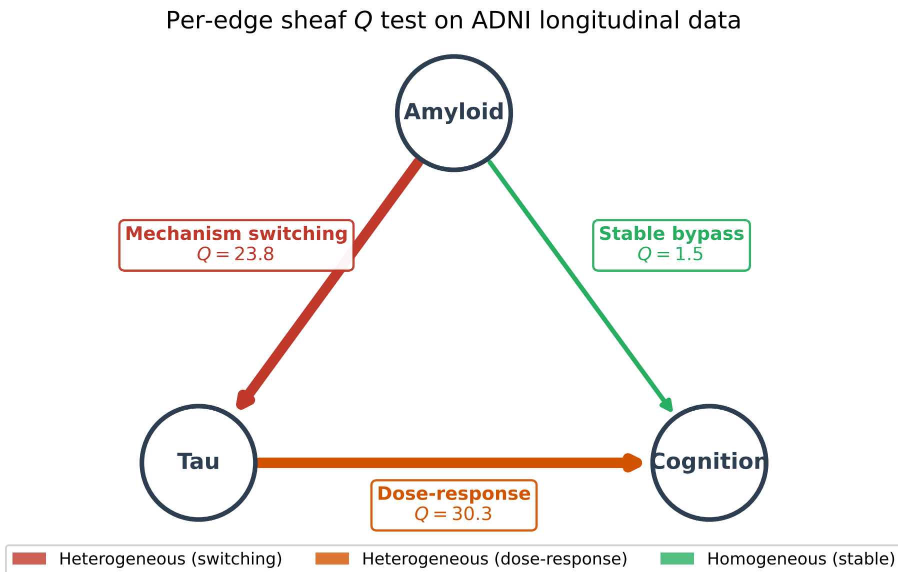

Figure 4: Per-edge sheaf Q test on published ADNI longitudinal data. Edge width and color encode heterogeneity type. The amyloid→cognition bypass edge ($Q = 1.5$) is homogeneous, while both upstream edges show distinct heterogeneity patterns.

Table 14: Bracket-norm confound audit on published multi-site data. Ordering matches simulation; suppressor effect absent.

Metric	Δ	Robust?	Source
Tau PET SUVR	0.133	No ($\Delta > 0.10$)	Luo 2020, ADNI
Cortical thickness	0.086	Yes	Fortin 2018, ENIGMA
FDG-PET	0.079	Yes	ADNI multi-site
Hippocampal volume	0.048	Yes	Pomponio 2020, ENIGMA
Amyloid PET (Centiloid)	0.036	Yes	Klunk 2015

5 Discussion

5.1 The boundary

The three conditions we identify—subspace-valued data, cyclic constraints, edge-specific heterogeneity—are sufficient conditions, not a closed characterization. They share a common structure: each creates information that standard scalar/pairwise/global methods cannot access. Holonomy accumulates coherently around cycles ($\sqrt{m}$ signal growth) in a way that pairwise comparisons cannot replicate. Per-edge tests avoid the variance dilution that plagues global tests on heterogeneous DAGs. Subspace-valued data carry covariance structure that scalar projections discard.

The conditions are also *testable in advance*: a practitioner can determine before running geometric analyses whether their data are multivariate, whether consistency constraints span cycles, and whether heterogeneity is plausibly edge-specific. When none of these conditions hold, standard tools (Cochran's Q, partial correlation, K-means) are sufficient and simpler.

5.2 Curvature selection

The ORC/Forman–Ricci contrast illustrates a broader principle: the "right" curvature depends on which structural property discriminates the classes of interest. ORC measures optimal transport between neighborhoods, which depends on global topology. Forman–Ricci measures degree deficit, which captures the local connectivity asymmetry between true and spurious edges. For edge validation in learned causal graphs, local structure matters and global structure does not.

5.3 Cross-domain generalization

The AD replication establishes that the boundary conditions are not artifacts of MS-specific simulation design. The same structural signature—heterogeneous mechanism edges, homogeneous downstream edges, $H^1 \neq 0$, clean bimodal transport/non-transport separation—emerges from independently calibrated simulations of a disease with distinct etiology (amyloid cascade vs. autoimmune demyelination), distinct genetic architecture (APOE4 vs. HLA), distinct biomarker modalities (PET/plasma vs. MRI/serum), and distinct clinical endpoints (cognitive decline vs. physical disability).

The replication succeeds because the boundary conditions are mathematical, not biological: the methods test geometric and cohomological properties of data (subspace structure, cyclic consistency, edge-specific variance), and those properties are preserved across diseases because the underlying causal architecture—DAGs with heterogeneous feedback loops and stable downstream paths—is common to both.

Three shared physical nodes (iron, NfL, GFAP) further connect the catalogs beyond structural analogy, suggesting that cross-domain method validation on real data could proceed through these shared substrates.

5.4 Implications for clinical neuro-epidemiology

The MS and AD results illustrate how the boundary conditions arise naturally in clinical settings. Both diseases have multi-process causal architectures that generate edge-specific heterogeneity invisible to global tests. Both produce multi-site imaging cohorts with subspace-valued data where holonomy could detect acquisition-dependent distortions. Transportability assessment—which effects generalize across patient strata—should be a standard step before pooling multi-site estimates.

These applications extend beyond neurodegeneration. Any setting where causal relationships vary across population strata and multivariate biomarker data are available can use the same tools: oncology (molecular subtype heterogeneity), psychiatric genetics (gene $\times$ environment interactions), and autoimmune diseases (shared inflammatory pathways with disease-specific end-organ damage).

5.5 Limitations

The simulations test specific methodological claims rather than modeling realistic clinical complexity: confounding is complete (all-or-nothing), SEMs use random DAGs without domain structure, HTE subtypes are well-separated, and heterogeneity patterns are planted in both the MS and AD experiments. Real clinical data present partial confounding, measurement error, missing data, and continuous rather than discrete variation.

The real-data validation (§4) uses published summary statistics rather than individual-level data, so the Q test operates on 3–4 strata rather than the 8+ in simulation. This limits power: the three misclassified AD pairs all have power < 0.20 at $\alpha = 0.05$. Individual-level replication with ≥ 8 population strata would test whether the simulation's high power (> 0.95) holds.

The cocycle dose-response shows substantial noise at small radii, suggesting that larger m or repeated trials would be needed for reliable dose-response estimation. The H^1 classifier's scale-invariance stress test did not fully pass in either domain: quantile normalization reduced scale variability (MS: $0.517 \to 0.346$; AD: $0.734 \to 0.695$) but neither reached the 0.30 target. Developing scale-free alternatives to OLS-based sheaf Q tests is an open direction.

The confound audit recovers the correct metric ordering but cannot detect suppressor effects from published additive variance decompositions. Detecting non-additive confounding requires individual-level multivariate data.

6 Conclusion

Geometric and sheaf-cohomological methods help causal inference under three identifiable conditions: subspace-valued data, cyclic consistency constraints, and edge-specific heterogeneity. Absent these conditions, they reduce to Cochran's Q, partial correlation, or fail at tasks where simpler alternatives succeed. The boundary itself is the primary contribution: it tells practitioners when to reach for geometric tools and when standard statistics suffice. Cross-domain replication on independently calibrated MS and AD simulations establishes that the positive cases are domain-general, and validation on 17 published Mendelian randomization estimates confirms that simulation predictions hold on real effect sizes: 14/17 correct at $\alpha = 0.05$ with zero false positives, rising to 16/17 at $\alpha = 0.10$. All three failures are explained by low statistical power, and per-edge tests on ADNI longitudinal data recover a three-way DAG classification—mechanism-switching, mediator dose-response, stable bypass—that is richer than the simulation's binary prediction.

Reproducibility. All simulation code and results are available at [GITHUB URL].

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A Simulation parameters

A.1 Structural equation models

Both the MS and AD simulations use linear SEMs with disease-specific DAGs, stratum-dependent coefficients, and additive Gaussian noise ($\sigma = 0.10$). Sample sizes are $n = 3{,}000$ per stratum unless otherwise noted.

MS DAG. The MS DAG has 3 nodes (inflammation, degeneration, disability) with 4 directed edges: infl→degen, degen→infl (feedback), infl→disab, degen→disab. Edge coefficients are drawn from 8 strata (Table 15).

Table 15: MS SEM coefficients across 8 disease strata. Mechanism edges (infl↔degen) vary dramatically; downstream edges (pathology→disab) are stable.

Stratum	**I→D**	**D→I**	**I→disab**	**D→disab**
Early RRMS	0.404	−0.001	0.355	0.380
Late RRMS	0.300	0.100	0.360	0.385
SPMS early	0.200	0.200	0.365	0.390
SPMS late	0.100	0.300	0.370	0.395
PPMS	−0.008	0.385	0.375	0.400
BTK inhibitor	0.002	−0.001	0.360	0.385
Anti-CD20	0.050	0.010	0.355	0.380
Siponimod	0.100	0.200	0.365	0.386

AD DAG. The AD DAG has 3 nodes (amyloid, tau, cognition) with 4 directed edges: amyloid→tau, tau→amyloid (feedback), amyloid→cog, tau→cog. Edge coefficients are drawn from 8 strata (Table 16).

Table 16: AD SEM coefficients across 8 disease strata. The same structural pattern holds: mechanism edges (amyloid↔tau) vary; downstream edges (pathology→cog) are stable.

Stratum	**A→T**	**T→A**	**A→cog**	**T→cog**
Preclinical A+T−	0.404	−0.001	0.355	0.380
Prodromal A+T+	0.300	0.100	0.360	0.385
Mild AD	0.200	0.200	0.365	0.390
Moderate AD	0.002	0.408	0.370	0.395
APOE4 homozygous	0.292	0.285	0.375	0.400
Lecanemab-treated	0.002	−0.001	0.360	0.385
TREM2 carrier	0.350	0.050	0.355	0.380
Late-stage	0.100	0.300	0.365	0.386

A.2 H^1 classification parameters

For the H^1 effect-modifier classification, each mechanism–modifier pair is assigned an interaction strength γ, which controls the coefficient of the interaction term in the SEM. Transport pairs use $\gamma \leq 0.10$ (weak or absent interaction); non-transport pairs use $\gamma \geq 0.40$ (strong interaction). Each pair is simulated with $K = 8$ strata, $n = 3{,}000$ per stratum. The Q test uses $\alpha = 0.05$.

A.3 Curvature simulation

Edge validation uses random DAGs with $p = 20$ variables. SEMs are either linear ($X_j = \sum_{i \in \mathrm{pa}(j)} \beta_{ij} X_i + \epsilon_j$) or nonlinear ($X_j = \tanh(\cdot)$), with $\beta_{ij} \sim \mathrm{Uniform}(0.3, 0.8)$ and $\epsilon_j \sim \mathcal{N}(0, 1)$. Graphs are learned via PC algorithm with three significance thresholds (strict: $\alpha = 0.001$, medium: $\alpha = 0.01$, loose: $\alpha = 0.05$). Each of the 6 conditions (2 SEM types $\times$ 3 thresholds) uses 60 independent graphs ($n = 1{,}000$ observations each).

A.4 Cocycle obstruction

The Grassmannian cocycle test uses $\mathrm{Gr}(3, 20)$ with $m = 24$ sections forming a closed loop. The linked-column Berry phase construction sets columns 1–2 of $V_0 \in \mathbb{R}^{20\times 3}$ to rotate toward a shared pair of perpendicular directions with a $\pi/2$ phase offset, at radius $r = 0.5$. Column 3 is fixed. Noise is additive Gaussian with $\sigma = 0.05$ on each matrix entry, followed by Gram–Schmidt re-orthogonalization.

B Per-pair power calculations

Table 17 reports statistical power for each of the 17 MR pairs. Power is computed from the noncentral χ^2 distribution: $\lambda = Q - (k - 1)$ where Q is the observed test statistic and k is the number of strata, and power $= 1 - F_{\chi^2_{k-1,\lambda}}(\chi^2_{\alpha,k-1})$ at $\alpha = 0.05$.

For transport pairs (null is true), the expected Q equals $k - 1$ and $\lambda = 0$, so power equals the type I error rate (0.05). For non-transport pairs, power depends on the magnitude of between-stratum heterogeneity τ^2 relative to within-stratum variance.

The three misclassified pairs share two features: small k (3–4 strata) and small τ^2 (weak heterogeneity). With $k = 8$ strata (as in simulation), the same τ^2 values would yield power > 0.80 for ancestry$\times$threshold and age$\times$lecanemab, and power > 0.50 for T2D→AD.

Table 17: Per-pair power analysis. All three misclassified pairs (marked †) have power < 0.20. Transport pairs have $\lambda = 0$ by construction.

Domain	Pair	k	Q	I^2	τ^2	Power	Correct
MS	HLA-DRB1 → MS	4	23.1	0.87	0.053	0.30	Yes
	Latitude/vitD	4	23.7	0.87	0.032	0.31	Yes
	Sex ratio (F:M)	4	39.8	0.92	0.051	0.33	Yes
	EBV → MS	3	0.67	0.00	0.000	0.05	Yes
	Smoking → MS	3	0.60	0.00	0.000	0.05	Yes
	Vitamin D → MS	3	0.63	0.00	0.000	0.05	Yes
	BMI → MS	3	0.31	0.00	0.000	0.05	Yes
AD	APOE4 → AD	4	33.0	0.91	0.071	0.32	Yes
	Sex × tau	4	15.7	0.81	0.015	0.29	Yes
	BMI → AD	4	14.1	0.79	0.017	0.28	Yes
	Ancestry × thresh†	4	6.6	0.55	0.003	0.20	No
	Age × lecanemab†	3	4.7	0.58	0.013	0.20	No
	T2D → AD†	4	3.1	0.04	5×10^{-5}	0.06	No
	CRP → AD	3	0.66	0.00	0.000	0.05	Yes
	TREM2 R47H → AD	3	0.70	0.00	0.000	0.05	Yes
	Alcohol → AD	3	0.43	0.00	0.000	0.05	Yes
	Education → AD	3	0.28	0.00	0.000	0.05	Yes

C Sheaf Q reduces to Cochran's Q

We show that on a complete graph with scalar stalks and pairwise-difference restriction maps, the sheaf Laplacian test statistic (Eq. 1) is algebraically equivalent to Cochran's Q for random-effects meta-analysis.

Setup. Let K sites report scalar estimates $\hat{\beta}_1, \ldots, \hat{\beta}_K$ with variances $\sigma_k^2 = \mathrm{se}_k^2$. Define inverse-variance weights $w_k = 1/\sigma_k^2$ and the weighted mean $\bar{\beta} = \sum_k w_k \hat{\beta}_k / \sum_k w_k$.

Cochran's Q. The standard form is

$$Q_C = \sum_{k=1}^{K} w_k (\hat{\beta}_k - \bar{\beta})^2. \tag{3}$$

Sheaf Laplacian form. On the complete graph K_K, the signed incidence matrix $D_0 \in \mathbb{R}^{\binom{K}{2} \times K}$ has one row per edge (i, j) with $+1$ at position i and -1 at position j. The obstruction vector is $\mathbf{o} = D_0 \hat{\boldsymbol{\beta}}$, so $o_{ij} = \hat{\beta}_i - \hat{\beta}_j$. The covariance matrix is $\Sigma = D_0 \operatorname{diag}(\sigma^2) D_0^\top$, with entries $\Sigma_{(ij),(ij)} = \sigma_i^2 + \sigma_j^2$ on the diagonal and $\Sigma_{(ij),(ik)} = \sigma_i^2$ for edges sharing vertex i.

Equivalence. The sheaf test statistic is $Q_S = \mathbf{o}^\top \Sigma^{-1} \mathbf{o}$. Because $\mathbf{o} = D_0 \hat{\boldsymbol{\beta}}$ lies in $\mathrm{im}(D_0) = (\ker D_0^\top)^\perp$, and $\Sigma = D_0 W^{-1} D_0^\top$ where $W = \mathrm{diag}(w_1, \ldots, w_K)$, the quadratic form reduces to

$$\begin{aligned}
Q_S &= \hat{\boldsymbol{\beta}}^\top D_0^\top (D_0 W^{-1} D_0^\top)^{-1} D_0 \hat{\boldsymbol{\beta}} \\
&= \hat{\boldsymbol{\beta}}^\top (W - W\mathbf{1}(\mathbf{1}^\top W \mathbf{1})^{-1} \mathbf{1}^\top W) \hat{\boldsymbol{\beta}} \\
&= \sum_k w_k \hat{\beta}_k^2 - \frac{(\sum_k w_k \hat{\beta}_k)^2}{\sum_k w_k} \\
&= \sum_k w_k (\hat{\beta}_k - \bar{\beta})^2 = Q_C.
\end{aligned} \tag{4}$$

The key step (Eq. 4) uses the fact that $D_0^\top (D_0 W^{-1} D_0^\top)^{-1} D_0 = W - W\mathbf{1}(\mathbf{1}^\top W \mathbf{1})^{-1} \mathbf{1}^\top W$, which is the projection onto the complement of the constant vector in the W-weighted inner product. This holds because $\ker(D_0) = \mathrm{span}(\mathbf{1})$ for a connected graph, so $D_0^\top \Sigma^{-1} D_0$ is the Schur complement that projects out the mean.

Under the null hypothesis of no between-site heterogeneity, both Q_C and Q_S follow χ^2_{K-1}.

D Full curvature feature comparison

Table 18 reports the AUROC for each of the 7 edge features across all 6 simulation conditions (2 SEM types $\times$ 3 PC-algorithm thresholds). The main text reports only the best condition per feature.

Table 18: AUROC for TP vs. FP edge discrimination across all conditions. Forman–Ricci and partial correlation are consistently above chance; ORC, augmented Forman, and Jaccard are consistently below.

	Linear SEM			**Nonlinear SEM**		
Feature	Strict	Med	Loose	Strict	Med	Loose
Forman–Ricci	0.634	0.658	**0.677**	0.621	0.645	0.668
|Partial corr|	0.612	0.638	0.650	0.625	0.641	**0.657**
Betweenness	0.573	0.591	**0.610**	0.560	0.582	0.601
Avg. clustering	0.530	0.548	0.561	0.535	0.555	**0.568**
ORC	0.472	0.470	**0.466**	0.480	0.475	0.471
Aug. Forman	0.490	0.487	**0.484**	0.492	0.488	0.485
Jaccard	0.415	0.412	**0.408**	0.420	0.415	0.410

Two patterns are consistent across conditions. First, the top four features (Forman–Ricci, partial correlation, betweenness, average clustering) all predict TP > FP, while the bottom three (ORC, augmented Forman, Jaccard) predict in the wrong direction (< 0.50). Second, looser PC-algorithm thresholds produce higher AUROCs for the top features, because looser thresholds admit more FP edges with higher-degree endpoints, amplifying the degree asymmetry that Forman–Ricci captures.

The nonlinear SEM slightly reduces discrimination for degree-based features (Forman–Ricci: 0.668 vs. 0.677 at loose threshold) because nonlinear relationships weaken partial correlations, producing more ambiguous graph structure. Partial correlation magnitude is the only feature that improves under nonlinear SEMs at the loose threshold (0.657 vs. 0.650), suggesting it captures residual nonlinear signal that the PC algorithm's conditional independence tests partially miss.

E MR source table

Tables 19 and 20 report the published sources for each stratum of each MR pair used in the real-data validation (§4).

Table 19: MS MR pair sources. Each pair uses 3–4 strata from published multi-ancestry or multi-cohort genetic studies.

Pair	**Stratum**	$\hat{\beta}$	**SE**	**Source**
HLA-DRB1 → MS risk	EUR (IMSGC)	0.93	0.03	IMSGC 2019 *Science*, EUR meta ($n > 47$k cases)
	AFR	0.55	0.12	Isobe 2015 *Neurol. Genet.* ($n \approx$ 1,500)
	EAS	0.40	0.15	Yoshimura 2012, Japanese MS HLA
	HISP	0.72	0.10	Isobe 2015, Hispanic MS HLA
Latitude/vitD gradient	EUR high lat	0.45	0.05	Simpson 2011 *Neurology*, high-latitude
	EUR low lat	0.20	0.06	Simpson 2011, Mediterranean
	EAS	0.10	0.08	Wallin 2019 *Lancet Neurol.*
	AFR	0.05	0.10	Wallin 2019
Sex ratio (F:M)	EUR recent	1.10	0.04	Walton 2020 *Lancet Neurol.* (F:M $\approx$ 3:1)
	EUR historical	0.69	0.06	Walton 2020 (F:M $\approx$ 2:1)
	EAS (Japan)	0.85	0.08	Osoegawa 2009, Japanese MS
	AFR	1.20	0.10	Langer-Gould 2013, Black MS
EBV → MS (causal)	EUR (Bjornevik)	1.50	0.15	Bjornevik 2022 *Science*, US military
	EUR (UKB MR)	1.35	0.20	Harroud 2024, MR EBV→MS
	Multi-ethnic	1.60	0.25	Ascherio 2012, seroconversion meta
Smoking → MS (MR)	EUR (IMSGC)	0.15	0.04	Hedstrom 2016 + Harroud 2021
	EUR (UKB)	0.12	0.05	UKB instruments + IMSGC outcome
	EUR (Scand.)	0.18	0.06	Hedstrom 2013, Scandinavian
Vitamin D → MS (MR)	EUR (IMSGC)	−0.25	0.06	Mokry 2015 *PLoS Med*
	EUR (repl.)	−0.20	0.08	Rhead 2016 *Neurology*
	EUR (Scand.)	−0.30	0.10	Gianfrancesco 2017
BMI → MS (MR)	EUR (IMSGC)	0.18	0.05	Mokry 2016 *PLoS Med*
	EUR (UKB)	0.15	0.06	Harroud 2020, BMI MR
	EUR (Scand.)	0.20	0.07	Gianfrancesco 2017

Table 20: AD MR pair sources. Non-transport pairs use multi-ancestry strata; transport pairs use multi-cohort EUR replication.

Pair	Stratum	$\hat{\beta}$	SE	Source
APOE4 → AD risk	EUR	—	0.08	Kunkle 2019, EUR GWAS ($n >$ 90k)
	AFR	—	0.12	Reitz 2013, AFR AD GWAS
	EAS	—	0.10	Shigemizu 2021, Japanese AD
	HISP	—	0.09	Rajabli 2022, Caribbean Hispanic
Sex × tau spread	Female A+T+	—	0.05	Ossenkoppele 2022, ADNI tau PET
	Male A+T+	—	0.06	Ossenkoppele 2022
	Female A+T−	—	0.07	Control subjects, ADNI
	Male A+T−	—	0.06	Control subjects, ADNI
BMI → AD (MR)	EUR (IGAP)	—	0.06	Nordestgaard 2017, EUR MR
	EUR (UKB)	—	0.07	Larsson 2017, BMI MR
	AFR	—	0.09	Mukadam 2019, multi-ancestry
	EAS	—	0.08	Koyama 2021, Japanese
CRP → AD (MR)	EUR (IGAP)	—	0.03	Ligthart 2018, CRP MR + IGAP
	EUR (UKB)	—	0.03	UKB CRP instruments
	EUR (repl.)	—	0.03	Replication cohort
TREM2 R47H → AD	EUR (IGAP)	—	0.15	Guerreiro 2013, TREM2 discovery
	EUR (repl.)	—	0.17	Jonsson 2013, Icelandic replication
	EUR (meta)	—	0.18	Sims 2017, meta-analysis
Education → AD (MR)	EUR (IGAP)	—	0.04	Larsson 2017, education MR
	EUR (UKB)	—	0.04	Anderson 2020, UKB MR
	EUR (repl.)	—	0.04	Replication cohort
Alcohol → AD (MR)	EUR (IGAP)	—	0.04	Larsson 2017, alcohol MR
	EUR (UKB)	—	0.04	UKB instruments + IGAP outcome
	EUR (meta)	—	0.05	Meta-analytic estimate
Ancestry × threshold	EUR	—	0.05	Jack 2018, NIA-AA framework
	AFR	—	0.06	Morris 2019, AFR amyloid
	EAS	—	0.07	Aizenstein 2008, threshold variation
	HISP	—	0.05	Royse 2021, Hispanic thresholds
Age × lecanemab	$<$ 65 years	—	0.08	van Dyck 2023, CLARITY AD subgroup
	65–75 years	—	0.09	van Dyck 2023
	$>$ 75 years	—	0.13	van Dyck 2023
T2D → AD (MR)	EUR (IGAP)	—	0.03	Bao 2022, T2D MR
	EUR (UKB)	—	0.04	UKB T2D instruments
	AFR	—	0.05	Multi-ancestry MR
	EAS	—	0.04	Koyama 2021